

Finite Fourier transform of doubly singular binary quintic forms

Mohamed Wafik

Advised by Dr. Frank Thorne

Department of Mathematics
McCausland College of Arts and Sciences
University of South Carolina

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Definition

Let $V_n(R)$ be the space of binary n -ic forms over a ring R ,

$$V_n(R) := \{f(x, y) = a_0x^n + a_1x^{n-1}y + \cdots + a_ny^n \mid a_0, a_1, \dots, a_n \in R\}.$$

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Let $G := \mathrm{GL}_1 \times \mathrm{GL}_2$ act by

$$\left(t, \begin{pmatrix} a & b \\ c & d \end{pmatrix} \right) \circ f(x, y) = \frac{t}{(ad - bc)^{\lfloor n/2 \rfloor}} f(ax + cy, bx + dy),$$

and pair V_n with itself by $[f, g] := \sum_{i=0}^n \frac{a_i b_i}{\binom{n}{i}}$, so that

$[f, h \circ g] = [h^{-T} \circ f, g]$; this identifies $V_n^* \cong V_n$ equivariantly.

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Definition

The **Fourier transform** of $\Phi : V_n \rightarrow \mathbb{C}$ is $\hat{\Phi} : V_n \rightarrow \mathbb{C}$,

$$\hat{\Phi}(f) := \frac{1}{p^{n+1}} \sum_{g \in V_n} \Phi(g) e^{2\pi i [f, g]/p}.$$

Why compute these transforms?

Explicit Fourier transforms of orbit indicator functions are the engine behind equidistribution and error-term results. Let $\omega_p : V_3 \rightarrow \mathbb{Z}_{\geq 0}$ count the distinct roots modulo p of a binary cubic form.

Theorem (Bhargava, Shankar and Tsimerman [BST])

Let $B \subseteq V_3$ be a box of side length $N \leq p$. Then

$$\sum_{f \in B} \omega_p(f) = N^4 + O(p^{3+\epsilon}).$$

The proof is Fourier inversion applied to $\omega_p = \sum_i c_i e_i$, using an *explicit* formula for each $\widehat{e_i}$.

Known results: quadratic forms

For $n = 2$ there are 4 orbits, one per splitting type:

$O_0 = \{0\}$, O_{1^2} , O_{11} , O_2 . Writing $e_1, \dots, e_4$ for their indicator functions, Taniguchi and Thorne [TT] proved

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$$\begin{pmatrix} \hat{e}_1 \\ \hat{e}_2 \\ \hat{e}_3 \\ \hat{e}_4 \end{pmatrix} = \frac{1}{p^3} \begin{pmatrix} 1 & 1 & 1 & 1 \\ p^2 - 1 & -1 & p - 1 & -(p + 1) \\ \frac{1}{2}p(p^2 - 1) & \frac{1}{2}p(p - 1) & -p & 0 \\ \frac{1}{2}p(p - 1)^2 & -\frac{1}{2}p(p - 1) & 0 & p \end{pmatrix} \begin{pmatrix} e_1 \\ e_2 \\ e_3 \\ e_4 \end{pmatrix}$$

Known results: cubic forms

With $O_0, O_{1^3}, O_{1^2 1}, O_{111}, O_{21}$ and O_3 , a similar identity was proven earlier by Mori [M] and using different methods by Taniguchi and Thorne [TT].

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$$\begin{pmatrix} \hat{e}_1 \\ \hat{e}_2 \\ \hat{e}_3 \\ \hat{e}_4 \\ \hat{e}_5 \\ \hat{e}_6 \end{pmatrix} = \frac{1}{p^4} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ p^2 - 1 & -1 & p - 1 & -1 + 2p & -1 & -1 - p \\ p(p^2 - 1) & (p - 1)p & (p - 2)p & -3p & -p & 0 \\ \frac{(p-1)^2 p(p+1)}{6} & \frac{(p-1)p(-1+2p)}{6} & -\frac{(p-1)p}{2} & \frac{p(\pm p+5)}{6} & -\frac{(\pm p-1)p}{6} & \frac{(\pm p-1)p}{6} \\ \frac{(p-1)^2 p(p+1)}{2} & -\frac{(p-1)p}{2} & -\frac{(p-1)p}{2} & -\frac{(\pm p-1)p}{2} & \frac{p(\pm p+1)}{2} & -\frac{(\pm p-1)p}{2} \\ \frac{(p-1)^2 p(p+1)}{3} & -\frac{p(p^2-1)}{3} & 0 & \frac{(\pm p-1)p}{3} & -\frac{(\pm p-1)p}{3} & \frac{p(\pm p+2)}{3} \end{pmatrix} \begin{pmatrix} e_1 \\ e_2 \\ e_3 \\ e_4 \\ e_5 \\ e_6 \end{pmatrix}$$

where $p \equiv \pm 1 \pmod{3}$.

For $n \leq 3$, every entry is a polynomial in p .

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Theorem (Ishitsuka, Taniguchi, Thorne and Xiao [ITTX])

For $p > 3$,

$$\widehat{\mathbb{1}_{\text{Disc}}}(f) = \begin{cases} p^{-1} + p^{-2} - p^{-3} & f = 0 \\ p^{-2} - p^{-3} & f \text{ has a triple root} \\ \chi_{12}(p)(-p^{-3} + p^{-4}) & f \in (1^2 1^2) \\ \chi_{12}(p)(p^{-3} + p^{-4}) & f \in (2^2) \\ \chi_{12}(p)p^{-4} & f \in (1^2 11) \text{ or } (1^2 2) \\ \left(\frac{-3I(f)}{p}\right) p^{-4} & J(f) = 0 \neq I(f) \\ a_p(E_f)p^{-4} & J(f) \neq 0 \neq \text{Disc}(f). \end{cases}$$

I, J are the quartic invariants, $\chi_{12} = \begin{pmatrix} 3 \\ \cdot \end{pmatrix}$, and $a_p(E_f) = p + 1 - |E_f(\mathbb{F}_p)|$ for $E_f : y^2 = x^3 - 3I(f)x^2 + J(f)^2$. **New feature: the last case is not a polynomial in p .**

The quintic question

A binary quintic form is **doubly singular** if it has at least 2 repeated roots, or a root of multiplicity at least 3.

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Can we compute $\widehat{\mathbb{1}_3}$ explicitly?

Independently and concurrently, Ishitsuka [1] computed $\widehat{\mathbb{1}_4}$ for singular quintic forms, i.e. with at most four distinct roots.

Sylvester canonical form

Over $\overline{\mathbb{F}_p}$ there exist linear forms l_1, l_2, l_3 (possibly zero or proportional) such that a binary quintic f takes one of the shapes

$$f = l_1^5 + l_2^5 + l_3^5, \quad f = l_1^5 + l_2^4 l_3, \quad f = l_1^3 l_2 l_3.$$

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With Abdesselam's [A] covariants $C_1 = (f, f)_4$, $C_2 = (f, C_1)_2$ and invariants J, K, L of degrees 4, 8, 12:

shape of f	detected by	notes
$l_1^3 l_2 l_3$	$J = K = L = 0$	$C_2(f, x, y)^2 = c_1 \operatorname{Disc}(l_1 l_2)^3 \operatorname{Disc}(l_1 l_3)^3 l_1^6$
$l_1^5 + l_2^4 l_3$	$L = 0$	$C_2(f, x, y) = c_1 \operatorname{Disc}(l_1 l_2)^2 \operatorname{Disc}(l_2 l_3) l_1 l_2^2$ $J(f) = \operatorname{Disc}(l_1 l_2)^4 \operatorname{Disc}(l_1 l_3)$ $K(f)^2 = c_2 \operatorname{Disc}(l_1 l_2)^{15} \operatorname{Disc}(l_2 l_3)^5$
$l_1^5 + l_2^5 + l_3^5$	$L \neq 0$	$C_2(f, x, y) = c_1 \operatorname{Disc}(l_1 l_2 l_3) l_1 l_2 l_3$

the roots of C_2 recover the Sylvester decomposition.

The arithmetic ingredients

For a binary quintic f over $\mathbb{F}_p$, $p > 5$:

- $\omega_p(f)$ = number of distinct roots of f in $\mathbb{P}^1(\mathbb{F}_p)$;
- $d_p(f)$ = number of distinct repeated roots of f in $\mathbb{P}^1(\mathbb{F}_p)$;
- $a_p(f) = p + 1 - |H_f(\mathbb{F}_p)|$, where H_f is the curve

$$H_f : z^2 = f_{xy}(x, y)^2 - f_{xx}(x, y) \cdot f_{yy}(x, y);$$

- with $\left(\frac{\cdot}{p}\right)$ the Legendre symbol,

$$s_p(f) := \sum_{[\alpha:\gamma] \in \mathbb{P}^1(\mathbb{F}_p), C_2(f, \alpha, \gamma)=0} \left(\frac{-2C_1(f, \alpha, \gamma)}{p} \right).$$

Main Theorem

Theorem (W.)

Let $p > 5$. If $J(f) = K(f) = L(f) = 0$ then

$$p^5 \widehat{\mathbb{1}}_3(f) = \begin{cases} 2p^3 - p & f = 0 \\ 2p^2 - 2p & f \in (1^5) \\ 2p^2 - 3p + 1 & f \in (1^4 1) \\ p^2 - 2p + 1 - (p-1) \left(\frac{6}{p}\right) & f \in (1^3 1^2) \\ p^2 - p + 2 + \left(\frac{6}{p}\right) & f \in (1^3 1 1) \\ p^2 - p + \left(\frac{6}{p}\right) & f \in (1^3 2), \end{cases}$$

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and otherwise

$$p^5 \widehat{\mathbb{I}}_3(f) = \begin{cases} -d_p(f)p + \omega_p(f) - 1 & L = K = 0 \neq J \\ (1 + s_p(f) - d_p(f))p + \omega_p(f) + a_p(f) - 1 & L = 0 \neq K \\ p \left[\left(\frac{L(f)}{p}\right) s_p(f) - d_p(f) \right] + \omega_p(f) + a_p(f) - 1 & L \neq 0. \end{cases}$$

Step 1: projectivize

The proof follows similar methods to those used in the literature for the singular binary quartic forms.

First, projectivizing the Fourier transform by writing

$$\begin{aligned} p^6 \widehat{\mathbb{I}}_3(f) &= \sum_{h \in V_5} \mathbb{I}_3(h) e^{2\pi i [h, f] / p} \\ &= 1 + \sum_{\substack{\bar{h} \in \mathbb{P}(V_5) \\ h \text{ is doubly singular for } h \in \bar{h}}} \sum_{h \in \bar{h}} e^{2\pi i [h, f] / p} \end{aligned}$$

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Step 2: counting X and X^f

$$X = \{\bar{h} \in \mathbb{P}(V_5) : h \text{ doubly singular}\}, \quad X^f = \{\bar{h} \in X : [h, f] = 0\}.$$

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Every $\bar{h} \in X$ is l^3q or lq^2 (l linear, q quadratic). Avoiding double counting in lq^2 by making sure q is square-free and $l \nmid q$, we get

$$|X| = \frac{p^2 - 1}{p - 1} \cdot \frac{p^3 - 1}{p - 1} + \frac{p^2 - 1}{p - 1} \left(\frac{p^2 - p}{2} + \frac{p(p - 1)}{2} \right) = 2p^3 + 2p^2 + p + 1.$$

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For X^f we parameterize by five families:

$$X_{1^5}^f = \{l \in \mathbb{P}(V_1) : [l^5, f] = 0\}, \quad X_{1^4 1}^f = \{(h, l_2) \in \mathbb{P}(V_1)^2 : [l_1^4 l_2, f] = 0\},$$

$$X_{1^3 2}^f = \{(l, q) \in \mathbb{P}(V_1) \times \mathbb{P}(V_2) : [l^3 q, f] = 0\},$$

$$X_{1^2 2^2}^f = \{(l, q) \in \mathbb{P}(V_1) \times \mathbb{P}(V_2) : [l q^2, f] = 0\},$$

$$X_{1^3 1^2}^f = \{(h, l_2) \in \mathbb{P}(V_1)^2 : [l_1^3 l_2^2, f] = 0\}.$$

Step 3: inclusion-exclusion

Each family maps onto part of X^f . Counting the fibres $\psi_{\bullet}^{-1}(\bar{h})$ over each splitting type gives the table

splitting type of $\bar{h}$	$ \psi_{1^5}^{-1}(\bar{h}) $	$ \psi_{1^4 1}^{-1}(\bar{h}) $	$ \psi_{1^3 2}^{-1}(\bar{h}) $	$ \psi_{1^2 2^2}^{-1}(\bar{h}) $	$ \psi_{1^3 1^2}^{-1}(\bar{h}) $
1^5	1	1	1	1	1
$1^4 1$	0	1	1	1	0
$1^3 1^2$	0	0	1	1	1
$1^2 2^2$	0	0	0	1	0
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$$\implies |X^f| = |X_{1^5}^f| - |X_{1^4 1}^f| + |X_{1^3 2}^f| + |X_{1^2 2^2}^f| - |X_{1^3 1^2}^f|$$

The first three counts are elementary, $|X_{1^3 1^2}^f|$ produces $a_p(f)$ and $|X_{1^2 2^2}^f|$ produces $s_p(f)$.

The three elementary counts

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So, $|X_{1^5}^f| = \omega_p(f)$.

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$$[(s_0x + s_1y)^4(t_0x + t_1y), f(x, y)] = \frac{1}{5} (t_0f_x(s_0, s_1) + t_1f_y(s_0, s_1))$$

So, $|X_{1^41}^f| = d_p(f) \cdot (p+1) + (p+1 - d_p(f)) = p \cdot d_p(f) + p + 1$.

The three elementary counts

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So, $|X_{1^41}^f| = d_p(f) \cdot (p+1) + (p+1 - d_p(f)) = p \cdot d_p(f) + p + 1$.

3

$$[l^3q, f(x, y)] = \frac{1}{20} (t_0f_{xx}(s_0, s_1) + t_1f_{xy}(s_0, s_1) + t_2f_{yy}(s_0, s_1))$$

leaves $|X_{1^32}^f| = p^2t + (p+1)^2$ where t is the number of distinct repeated roots of f of multiplicity ≥ 3 modulo p .

$X_{1^3 1^2}^f$: where $a_p(f)$ comes from

$$20[(s_0x + s_1y)^3(t_0x + t_1y)^2, f(x, y)] = (t_0 \ t_1) \begin{pmatrix} f_{xx}(s_0, s_1) & f_{xy}(s_0, s_1) \\ f_{xy}(s_0, s_1) & f_{yy}(s_0, s_1) \end{pmatrix} \begin{pmatrix} t_0 \\ t_1 \end{pmatrix}.$$

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If the **Hessian** matrix is 0, then there are $p + 1$ points to count for $[t_0 : t_1]$. Otherwise, the binary quadratic form has a number of solutions equal to the count of solutions to

$$z^2 = -\det \begin{pmatrix} f_{xx}(s_0, s_1) & f_{xy}(s_0, s_1) \\ f_{xy}(s_0, s_1) & f_{yy}(s_0, s_1) \end{pmatrix}.$$

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So other than $f = l_1^3 l_2 l_3$, $|X_{1^3 1^2}^f| = |H_f(\mathbb{F}_p)|$, the curve

$$H_f : z^2 = f_{xy}(x, y)^2 - f_{xx}(x, y) \cdot f_{yy}(x, y).$$

$X_{12^2}^f$: where $s_p(f)$ comes from

$$[(s_0x + s_1y)(t_0x^2 + t_1xy + t_2y^2), f] = (t_0 \ t_1 \ t_2)M(s_0, s_1) \begin{pmatrix} t_0 \\ t_1 \\ t_2 \end{pmatrix}$$

where

$$|M(s_0, s_1)| = -\frac{1}{6} C_2(f, s_0, s_1)$$

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This can be viewed as a ternary quadratic form with

$p + 1$ points if $M(s_0, s_1)$ is of rank 1 or 3,

$2p + 1$ or 1 points if $M(s_0, s_1)$ is of rank 2 and

$p^2 + p + 1$ points if $M(s_0, s_1) = 0$.

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Thank you!

Backup: $|X_{1^3 1^2}^f|$ when $J = K = L = 0$

For $L(f) = K(f) = J(f) = 0 \neq f$,

$$|X_{1^3 1^2}^f| = \begin{cases} 2p + 1 & f \in (1^5) \\ 3p + 1 & f \in (1^4 1) \\ 2p + 1 + (p - 1) \left(\frac{6}{p} \right) & f \in (1^3 1^2) \\ 2p + 1 - \left(\frac{6}{p} \right) & f \in (1^3 1 1) \text{ or } f \in (1^3 2) \end{cases} .$$

Otherwise, $|X_{1^3 1^2}^f| = |H_f(\mathbb{F}_p)|$.

Backup: $X_{12^2}^f$ for $f = l_1^3 l_2 l_3$

For $f = l_1^3 l_2 l_3$, we get $|M| = -\frac{1}{6} C_2(f, x, y)$ with $C_2(f, x, y)^2 = c_1 \text{Disc}(l_1 l_2)^3 \text{Disc}(l_1 l_3)^3 l_1^6$ for some $c_1 \in \mathbb{F}_p^\times$, and there are 2 separate cases:

$|M|$ or C_2 is identically 0 with $f = l_1^4 l_2$, or
not identically 0 with $f = l_1^3 l_2 l_3$ with $l_2 \neq l_1 \neq l_3$.

- ① If $C_2(f, x, y)$ is identically 0, then we have $(\mathbf{p} + \mathbf{1})^2 + \mathbf{p}^2$ points.
- ② If $C_2(f, x, y)$ is not identically 0, then we have $(\mathbf{p} + \mathbf{1})^2 + \mathbf{p}$ points.

Backup: $X_{12^2}^f$ for $f = l_1^5 + l_2^4 l_3$ and $f = l_1^5 + l_2^5 + l_3^5$

- ① $l_2 = l_3$, so $K(f) = 0 \neq J(f)$: averaging over all $[s_0 : s_1]$ yields $(\mathbf{p} + \mathbf{1})^2$ points.
- ② $l_1 = l_3$, so $K(f) \neq 0 = J(f)$: we have $g \circ f = ax^5 + bxy^4$ for some $g \in \text{GL}_2(\mathbb{F}_p)$, with dependency on $-5ab$, and $\left(\frac{-5ab}{p}\right) = \left(\frac{-2C_1(f, \alpha, \gamma)}{p}\right)$ where $[\alpha : \gamma]$ is the root of $C_2(f, x, y)$ that is not also a root of $C_1(f, x, y)$. We end up with $(\mathbf{p} + \mathbf{1})^2 + \mathbf{p}[\mathbf{1} + \mathbf{s}_p(\mathbf{f})]$.
- ③ $l_1 \neq l_3 \neq l_2$, so $K(f) \neq 0 \neq J(f)$: again $(\mathbf{p} + \mathbf{1})^2 + \mathbf{p}[\mathbf{1} + \mathbf{s}_p(\mathbf{f})]$.

Last case, $L(f) \neq 0$:

$$(\mathbf{p} + \mathbf{1})^2 + \mathbf{p} \left[\sum_{[\alpha:\gamma] \in \mathbb{P}^1(\mathbb{F}_p), C_2(f, \alpha, \gamma) \neq 0} \left(\frac{-2L(f)C_1(f, \alpha, \gamma)}{p} \right) \right]$$